12.359

Kavin B-EE25BTECH11033

October 4,2025

Question

Which one of the following statements is NOT CORRECT?

1. A well-conditioned matrix has a condition number close to 1
2. An ill-conditioned matrix has a large condition number
3. The inverse of a well-conditioned matrix can be computed accurately
4. A non-invertible matrix has a condition number close to 1

Theoretical Solution

The **condition number** of an invertible matrix $\mathbf{A}$, denoted as $\kappa(\mathbf{A})$, is defined as:

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \left\|\mathbf{A}^{-1}\right\| \tag{1}$$

It measures how much the output of a linear system $\mathbf{Ax} = b$ can change for a small change in the input data.

Theoretical Solution

1) **A well-conditioned matrix has a condition number close to 1:** This is **CORRECT**. The smallest possible condition number is 1 (for example, the identity matrix). A value close to 1 indicates that the matrix is numerically stable.
2) **An ill-conditioned matrix has a large condition number:** This is **CORRECT**. This is the definition of an ill-conditioned matrix. A large condition number means that small errors in the input can lead to very large errors in the solution.

Theoretical Solution

3) **The inverse of a well-conditioned matrix can be computed accurately:** This is **CORRECT**. Because a well-conditioned matrix is not sensitive to small perturbations, numerical algorithms used to compute its inverse will produce results with high accuracy.

4) **A non-invertible matrix has a condition number close to 1:** This is **NOT CORRECT**. A non-invertible (or singular) matrix does not have an inverse $\mathbf{A}^{-1}$. For the purposes of conditioning, its inverse is considered to have an infinite norm. Therefore, the condition number of a non-invertible matrix is defined to be **infinity (∞)**. An infinite condition number is the largest possible value, representing the worst possible conditioning, not the best (which is close to 1).